EE-S-321F
March 22, 1982
SUPERSEDING
Fed. Spec. EE-S-321E
January 16, 1975

FEDERAL SPECIFICATION

SHORTENING COMPOUNDS AND LARD

This specification was approved by the U.S. Department of Agriculture for use by all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for shortening compounds and lard for the use of all Federal agencies.

1.2 Classification.

1.2.1 Types and classes. Shortening shall be of the following types and classes as specified (see 6.2):

Type I - Lard

Class 1 - Refined
Class 2 - Service style

Type II - General purpose shortening

Class 1 - Regular

(a) Plasticized (animal/vegetable fats or oils)
(b) Plasticized (all vegetable oil)

Class 2 - High stability (plasticized)

(a) Animal/vegetable fats or oils
(b) Cottonseed oil/peanut oil
(c) Soybean oil

Type III - Bakery (emulsifier type) shortening

Class 1 - High monoglyceride level

(a) Deodorized, stabilized lard and other deodorized shortenings
(b) Undeodorized, stabilized lard

FSC 8945

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Class 2 - Low monoglyceride level or other equivalent emulsifiers

(a) Plasticized
(b) Fluid

Type IV - Deep-fry cookery shortening

Class 1 - Plasticized
Class 2 - Fluid

(a) Animal/vegetable fats or oils
(b) All vegetable oil

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
TT-C-495	-	Coatings, Exterior for Tinned Food Cans
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-29	-	Canned Subsistence Items, Packaging and Packing Of
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

Federal Standard:

FED-STD-123	-	Marking for Shipment (Civil agencies)

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

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(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-L-1497	-	Labeling of Metal Cans for Subsistence Items
MIL-L-35078	-	Loads, Unit: Preparation of Nonperishable Subsistence

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-668	-	Sanitary Standards for Food Plants

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

US Department of Agriculture

Regulations Governing the Meat Inspection of the US Department of Agriculture, (9 CFR Part 200 to end)

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

United States Standards for Condition of Food Containers

(Application for copies should be addressed to the U.S. Department of Agriculture, Agricultural Marketing Service, Food Quality Assurance Division, South Building, 14th and Independence Avenue, SW, Washington, DC 20250.)

US Department of Health and Human Services

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder, (21 CFR Parts 100-199)

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

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American Oil Chemists Society

Official and Tentative Methods of the American Oil Chemists Society

(Copies may be obtained from the American Oil Chemists Society, 508 South Sixth Street, Champaign, IL 60601.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Fats and oils. Lard shall be made from good quality, fresh, clean hog fat. Animal fats and oils used in the shortening shall also be fresh, sweet, clean, and of good quality. Vegetable fats and oils shall be of good quality and all fats and oils used shall be adequately processed to meet the finished product requirements of the type specified.

3.1.2 Antioxidants and synergists. The use of antioxidants or combinations of antioxidants and synergists for use with certain antioxidants shall, where applicable, be in accordance with the regulations of the Food Safety and Inspection Service (FSIS) of the United States Department of Agriculture, and Federal Food, Drug, and Cosmetic Act.

3.1.3 Emulsifying agents. Emulsifying agents shall be monoglyceride concentrates, or other suitable emulsifying agents which are permitted by the Federal Food, Drug, and Cosmetic Act or the Federal Meat Inspection Act, as applicable, for use in shortening compounds and lard.

3.1.4 Anti-foaming agents. The use of anti-foaming agents shall be in accordance with regulations of the Food and Drug Administration promulgated under the Food Additive Amendment or the Federal Meat Inspection Act, as applicable, concerning rendered animal fats, processed vegetable fat, or blends thereof.

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3.2 Preparation. All lard products shall be prepared only in a plant which is regularly operated under the supervision of the Food Safety and Inspection Service (FSIS), United States Department of Agriculture, and shall be inspected, and if passed, marked in accordance with the Regulations Governing Meat Inspection of the United States Department of Agriculture.

3.2.1 Type I, lard.

3.2.1.1 Class 1, refined. Class 1 product shall be refined and prepared from steam-rendered, open-kettle-rendered, dry-rendered, drip-rendered, or low-temperature-rendered lard or a combination of these and may contain added hydrogenated lard. Antioxidants shall be incorporated into the product. It shall be plasticized to a uniform texture as specified (see 6.2) by passing over a chilling roll or through a closed, continuous chilling unit.

3.2.1.2 Class 2, service style. Class 2 product shall conform to the requirements for class 1 refined, and in addition, shall contain not less than 4 percent hydrogenated lard flakes having a titer of not less than 57°C. Antioxidants shall be incorporated into the product.

3.2.2 Type II, general purpose.

3.2.2.1 Class 1a, regular. Class 1a regular product shall be prepared from animal or vegetable fats or oils or blends thereof. All or part of these fats or oils may be processed by interesterification or hydrogenation or both. All of the fats and oils shall be deodorized. Antioxidants may be incorporated into the product. Anti-foaming agents in amounts that will not hinder baking performance may be added.

3.2.2.2 Class 1b, regular. Class 1b product shall be prepared from hydrogenated deodorized 100 percent vegetable oils, or blends thereof. No antioxidants or additives shall be incorporated.

3.2.2.3 Class 2a, high stability, all ingredient. Class 2a product shall be prepared from animal or vegetable fats or oils or blends thereof. All or part of these oils may be processed by interesterification or hydrogenation or both. All of the fats and oils shall be deodorized. Antioxidants may be incorporated into the product. The finished product shall have increased AOM stability as specified in table I.

3.2.2.4 Class 2b, high stability. Class 2b product shall be prepared from hydrogenated cottonseed or peanut oils, or blends thereof. Antioxidants may be added. The finished product shall have increased AOM stability as specified in table I.

3.2.2.5 Class 2c, high stability. Class 2c product shall be prepared from hydrogenated soybean oil with the addition of hydrogenated cottonseed oil or other suitable hydrogenated vegetable oil as required to conform to the plastic texture uniformity requirements contained in 3.3.1.2. The linolenic acid content of the finished product shall not exceed 3.0 percent. Antioxidants may be added. The finished product shall have increased AOM stability as specified in table I.

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3.2.3 Type III, bakery (emulsifier type).

3.2.3.1 Class 1a, b, and class 2a, b. Class 1a and 2a products shall be prepared from animal or vegetable fats and oils or blends thereof. All or part of these fats or oils may be processed by interesterification or hydrogenation or both. All of the fats and oils shall be deodorized. Class 1b shall be prepared from undeodorized stabilized lard. The undeodorized lard shall be free from cracklings, burnt fat, black particles, etc. Class 2b product shall be prepared only from vegetable fats and oils and shall be manufactured and packaged not more than 90 days before shipment. Antioxidants may be incorporated in all classes of type III products. Anti-foaming agents shall not be added. All classes of type III product except class 2b shall contain monoglyceride levels as specified in table I. Class 2b may contain any appropriate emulsifier system which is in compliance with 3.5.

3.2.4 Type IV, deep fry cookery.

3.2.4.1 Class 1, plasticized. Class 1, plasticized product, shall be prepared from animal or vegetable fats or oils or blends thereof. Coconut oil shall not be used. All or part of these fats or oils may be processed by interesterification or hydrogenation. All the fats and oils shall be deodorized. Anti-foaming agents shall be incorporated. Antioxidants may be incorporated.

3.2.4.2 Class 2, fluid. Class 2(a) fluid product, shall be prepared from fats and oils specified in 3.2.4.1. Class 2(b) shall be prepared from 100 percent vegetable oils or blends thereof. Anti-foaming agents shall be incorporated. Antioxidants may be incorporated.

3.3 Finished product.

3.3.1 Physical characteristics.

3.3.1.1 Type I lard. Type I product shall have a fresh, sweet, clean flavor and odor, free from rancidity, sourness, staginess or other objectionable flavors and odors after being subjected to the heating test specified in 4.4.2. The product shall have a uniform smooth texture, if grainy texture is not specified (see 6.2).

3.3.1.2 Types II, III, and IV, shortenings. Types II, III, and IV shortenings shall be free from rancidity, sourness, or other objectionable flavors and odors. Type II and IV products shall be checked for flavor and odor as described in 4.4.2 after being subjected to the heating test. Type III product shall be white and shall be checked for flavor and odor without subjecting product to the heating test of 4.4.2. Types II, III, and IV products shall have a uniform plastic texture, workable over a temperature range of 65° to 95°F with the exception of type IV, class 2, which shall be homogenous in fluidity so that shaking or stirring will not be necessary before using. Type III, class 2b may separate on standing and shall be capable of being made homogenous by shaking or agitation. Type III, class 2b and type IV, class 2 products shall pass the pouring test specified in 4.4.3. Type IV classes 1 and 2 shall pass the foam test specified in 4.4.4.

3.3.2 Analytical requirements. The finished product shall conform to the analytical requirements of table I.

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TABLE I. Analytical requirements

Types and classes	Free fatty acids (percent)	Peroxide value (meq of peroxide per kg of fat)	Wiley melting point (°F.)	Fat stability AOM (hours)	Smoke point (°F.)	Alpha monoglycerides (percent)	Moisture and volatile matter (percent)	Lovibond color (red)
	Max.	Max.	Range	Min.	Min.	Range	Max.	Max. 1/
Type I								
Class 1	0.5		96-106	40	340		0.20	2.5
Class 2	0.5		not less than 116	40	340		0.20	2.5
Type II								
Class 1a	0.10	0.5		40	400		0.10	2.5
Class 1b	0.05	1.0	112-120	75	425		0.10	2.5
Class 2a	0.08	0.5		100	400		0.10	2.5
Class 2b	0.08	0.5		100	400		0.10	2.5
Class 2c	0.08	0.5		100	400		0.10	2.5
Type III								
Class 1a	0.3	0.5		40		5.0 to 7.0	0.20 2/	2.5
Class 1b	0.6	2.5		40		5.0 to 7.0	0.20 2/	2.5
Class 2a	0.15	0.5		40		1.5 to 5.5	0.20 2/	2.5
Class 2b	1.6	2.0		15		(see 3.2.3.1)	0.20 2/	2.5
Type IV								
Class 1	0.05	0.5		40	400		0.10	2.5
Class 2	0.05	0.5		25	420		0.10	2.5

1/ A maximum of 12.0 red shall apply to specified shortenings possessing a characteristic color (see 6.2).

2/ Requirement applicable to moisture only. If shortening compounds or lard is to be packed in cans, the maximum moisture shall not exceed 0.07 percent.

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3.3.3 Solid fat index (SFI). The shortening shall have plasticity characteristics, as determined by the dilatometric measurement of the fats and oils, in accordance with table II.

TABLE II. Solid fat index

Types and classes	Temperature		
Type II	50°F	70°F	92°F
Class 1a	26-38	18-30	11-25
Class 1b	23-31	18-24	12-17
Class 2a	26-42	18-30	10-23
Class 2b	27-42	20-30	10-20
Class 2c	27-42	20-30	10-22
Type III			
Class 1a	20-35	15-27	6-18
Class 1b	20-35	15-27	6-18
Class 2a	26-38	18-30	11-25
Class 2b	5-20	5-20	0-20
Type IV			
Class 1	26-42	18-30	6-18
Class 2	0-11	0-8	0-6

3.4 Storage and delivery temperature. Product unit packed in polyethylene bag in fiberboard box as specified in 5.1.2 shall be stored, handled, and delivered at chill temperatures not to exceed 75°F (see 6.1).

3.5 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.6 Plant qualification.

3.6.1 For civil agencies. The processing plant, grounds, equipment, personnel practices, and sanitary practices used in the production shall be such as to minimize the possibility of contamination of the product through microbial growth, dust, pests, rodents, condensate, or other unsanitary sources. The product shall be prepared and processed with minimum delay between the various stages in production. The product shall be packaged under modern sanitary conditions into clean, dust-free, sanitary containers.

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3.6.2 For military agencies. The product containing only vegetable fats shall be processed in establishments meeting the requirements of MIL-STD-668. Product containing any animal fats shall be processed and shipped from establishments under continuous inspection supervision of the Food Safety and Inspection Service (FSIS), U.S. Department of Agriculture, as officially listed by that agency, or from establishments listed in the Directory of Sanitarily Approved Food Establishments for Armed Forces Procurement. Lard products shall be processed in accordance with 3.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Plant qualification. Product shall be unacceptable if produced in plants not meeting the requirements of 3.6.

4.3 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.1.1 Component examination. Conformance of components to identity requirements of 3.1 and 3.2 as concerns source, processing history and kinds of ingredients permitted shall be ascertained by examination of pertinent labels, invoices, or other valid documents. In addition, each component shall be examined to determine conformance to the condition requirements of 3.1. The sample unit shall be approximately 1 pound derived from each of five primary containers. In the event a lot consists of less than five containers, a sample unit shall be extracted from each container in the lot. The use of nonconforming components shall be cause for rejection of the involved finished product.

4.3.1.2 Component testing. In addition to any testing required by 4.3.1, components shall be tested in accordance with table III. Any test failure shall be cause for rejection of the related component lot. Use of such component shall be cause for rejection of the involved quantity of finished product.

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TABLE III. Component tests

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref. and rqmts. paragraph
Cans 1/	1 can (or 4 inches by 4 inches square) and lid	Cans	S-1	Tin plate (not applicable to cans constructed of CMQ)	Nearest 0.01 lb/base box	5.1.1 (as applicable), 5.1.2.2 and 4.4.5
				Base plate wt.	Nearest lb/base box	
Polyethylene bags 1/	1 bag	Bags	S-1	Thickness 2/	Nearest 0.001 inch	5.1.2.1 4.4.7
				Seal strength of seam	Nearest percent	5.1.2.1 4.4.7
Hydrogenated lard flakes	One pound composite 3/	Primary containers	S-3	Titer	Nearest 1°C	3.2.1.2 4.4.1
Plastic containers 1/	One container	Containers	S-1	Thickness 2/	Nearest 0.001 inch	5.1.2.3 4.4.8

1/ Applicable to lot average.

2/ In lieu of performing test for thickness, conformance to requirement may be ascertained by examination of labels, invoices or other valid documents.

3/ Extract one pound sample from each primary container. Requirements shall be applicable to the sample unit.

4.3.2 In-process inspection. Unless otherwise specified (see 6.1), examination shall be made during processing to determine conformance with 3.2, as concerns procedures, equipment and formulation. Records of formulation shall be maintained. Nonconformance to one or more requirements as determined by actual examination or as reflected by the records shall be cause for rejection of the involved finished product.

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4.3.3 End item examination.

4.3.3.1 Can examination. The filled and sealed cans shall be examined for external conditions in accordance with the applicable requirements of the US Standards for Condition of Food Containers.

4.3.3.2 Polyethylene bag in fiberboard box examination. The filled and sealed polyethylene bags in fiberboard boxes shall be examined for the defects listed in table IV. The lot size shall be expressed in units of one bag in box. The sample unit shall be one bag in box as applicable. The inspection level shall be I and the acceptable quality level (AQL) shall be 1.5 major B and 2.5 minor defects per hundred units.

TABLE IV. Polyethylene bag in fiberboard box defects

Category		Defect	Polyethylene bag	Fiberboard box
Major B	Minor			
151		Not closed	-	X
152		Tear or hole	X	X
153		Evidence of grease seepage or staining	X	X
154		Top of bag not neatly folded over, protecting product from shipping container	X	-
155		National stock number, item description, contract number, or date of pack markings missing, incorrect or illegible	-	X
156		Not heat sealed along three edges	X	-
157		Height and width of bag in relation to fiberboard box not as specified	X	-
	201	Not clean	X	X
	202	Other required markings missing, incorrect, or illegible	-	X
	203	Thickness of bag not as specified	X	-

4.3.3.3 Plastic container examination. The filled and closed plastic containers shall be examined for the defects listed in table V. The lot size shall be expressed in units of one container. The sample unit shall be one container. The inspection level shall be I and the AQL shall be 0.25 major A and 1.5 major B defects per hundred units.

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TABLE V. Plastic container defects

Category		Defect
Major A	Major B	
101		Not material or style specified
	151	Evidence of leakage 1/
	152	Greaseproof liner missing from cap
	153	Cap not tight 2/
	154	Label missing, illegible, or incorrect

1/ To determine leakage, place the filled and closed container in an inverted position on a clean fiberboard pad. Hold in this position for a minimum of 30 seconds. Any evidence of oil stains on the fiberboard pad shall be considered a defect.

2/ To determine tightness of screwcap closure, place the plastic container on top of a filled and closed fiberboard container, grasp the sides of the cap firmly with fingers of one hand and turn cap counterclockwise with a minimum of downward pressure for one complete revolution. If cap turns and plastic container remains stationary, a defect shall be scored.

4.3.3.4 Net weight and volume examination. The primary containers shall be examined for the defects listed in table VI. The lot size shall be expressed in units of one container. The sample unit shall be one container. The inspection level shall be S-3 and the AQL shall be 2.5 percent defective. In addition the lot shall be rejected if the sample average is less than the specified net weight or volume.

TABLE VI. Net weight or volume defects 1/

Category	Defect
Minor	
201	Net weight or volume more than 2 percent under specified weight or volume (container size of 1 pound but not more than 8 pounds) 2/
202	Net weight or volume more than 1 percent under specified weight or volume (container size of more than 8 pounds) 3/

1/ Lot shall be rejected if sample data indicate lot average net weight is less than specified net weight.

2/ Containers 8 pounds or less shall be reported to the nearest 1/4 ounce, or 1 gallon or less to the nearest fluid ounce.

3/ Report to the nearest ounce or fluid ounce.

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4.3.3.5 Product examination. The product shall be examined for the defects listed in table VII. The lot size shall be expressed in units of one primary container. The sample unit shall be one primary container. For containers of less than 50 pounds, 1 pound of product shall be extracted for examination. For containers of 50 pounds or more, 5 pounds shall be extracted for examination. The inspection level shall be S-3 and the AQL shall be 1.5 major B defects per hundred units. The presence of any major A defect shall be cause for rejection of the lot.

TABLE VII. Product defects

Category		Defect
Major A	Major B	
101		Rancid, sour, staggy, or other objectionable flavor or odor, or type I product not white (when rated as specified in 4.4.1).
	151	Type I product does not have a uniform smooth texture, or does not have a grainy texture when grainy texture is specified
	152	Type II, III (except class 2b) or IV product (except type IV, class 2) does not have a uniform plastic texture workable between 65° to 95°F
	153	Type IV, class 2 product is not homogenous in fluidity (see 3.3.1.2)
102		Presence of foreign material (e.g., rust, dirt, wood particles, paint, insects, insect fragments, glass, hair, metal, etc.)

4.3.4 End item testing.

4.3.4.1 Product testing. The product shall be tested in accordance with table VIII. The lot size shall be expressed in units of one primary container. The sample unit shall be one primary container. For the pouring test, the inspection level shall be S-1. For all other tests, the sample shall be a composite derived from the number of primary containers indicated by inspection level S-2. For types I, II, and III, the sample shall be a 3-pound composite. For type IV, the sample shall be a 6-pound composite. Any test failure shall be cause for rejection of the lot.

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TABLE VIII. End item tests

Characteristic	Requirement paragraph	Test method	Results reported to nearest
Linolenic acid content (type II, class 2c)	3.2.2.5	4.4.1	0.1 percent
Pouring (type III, class 2b and type IV, class 2)	3.3.1.2	4.4.3	Pass or fail
Foam (type IV, classes 1 and 2)	3.3.1.2	4.4.4	Pass or fail
Analytical requirements	3.4	4.4.1	1/
Solid fat index (types II, III, and IV)	3.4.1	4.4.1	1 index number

1/ Report to same unit or decimal specified for the applicable requirement. In the event of failure for fat stability, report results to nearest 1 hour.

4.3.4.2 Can leakage testing. All types of filled and sealed cans shall be inspected for leakage in accordance with PPP-C-29 except that the leakage examination procedure for 1-gallon oblong cans and 5-gallon square cans shall be as specified in 4.4.6.

4.3.5 Packaging inspection.

4.3.5.1 Examination of fiberboard containers. When shipping containers are required to conform to PPP-B-636, the examination of the filled, closed, and reinforced (when applicable) shipping container shall be performed in accordance with that specification. In addition, the following defects shall be included in the table of examination, as applicable: Major: National stock number, item description, contract number or date of pack markings missing, incorrect, or illegible. Minor: (1) Other required markings missing, incorrect or illegible. (2) Arrangement or number of cans or plastic containers (as applicable) not as specified. (3) Full height partitions or pads missing or not as specified (when applicable). For level B packs, containers reinforced with other than nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape intended for transfer at sea shall not be acceptable. Level C containers shall be examined for the marking defects specified above and for the closure method specified in 5.2.3.

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4.3.5.2 Examination of metal drums (level C pack). The filled and closed drums shall be examined to determine the presence of the following defects:

(1) Insecure or improper closure.
(2) Evidence of leakage, tear, or hole.
(3) Marked net weight in excess of 400 pounds.
(4) Major: National stock number, item description, contract number or date of pack markings missing, incorrect or illegible.
Minor: Other required markings missing, incorrect or illegible.

The sample unit shall be one filled and closed drum and the sample size shall be the number of sample units indicated by inspection level I. The AQL shall be 2.5 major B defects per hundred units. Lot size shall be expessed in units of one drum. In addition, finding of one or more drums not complying with the minimum Uniform Freight Classification or National Motor Freight Classification, as applicable, as reflected by manufacturer's compliance symbol, shall be cause for rejection of the lot.

4.3.5.3 Examination of 50-pound capacity cans. The filled and closed cans shall be examined to determine the presence of the following defects:

(1) Insecure or improper closure.
(2) Slip cover and can bottom not constructed as specified.
(3) Can size other than specified.
(4) Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible. Minor: Other required markings missing, incorrect, or illegible.

The sample unit shall be one filled and closed can and the sample size shall be the number of sample units indicated by inspection level I. The AQL shall be 2.5 major B defects per hundred units. Lot size shall be expressed in units of one can.

4.4 Methods of inspection.

4.4.1 Chemical analyses. The following chemical analyses shall be made in accordance with methods of the American Oil Chemists' Society, Section c, Commercial Fats and Oils.

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Test	Method
Finished product:	
Free fatty acids (FFA)	Ca 5a -40
Peroxide value (PV)	Cd 8 -53
Fat stability (AOM)	Cd 12 -57
Smoke point	Cc 9a -48
Melting point (WMP)	Cc 2 -38
Moisture and volatile matter (types I, II and IV)	Ca 2d -25
Moisture (type III)	Ca 2e -55
Linolenic acid	Ce 1 -62
Color	Cc 13b-45
Alpha-monoglyceride	Cd 11 -57
Solid fat index (SFI)	Cd 10 -57
Ingredient:	
Titer	Cc 12 -59

4.4.2 Heating procedure. A small sample of the finished product (approximately 50 grams) shall be placed in a clean glass beaker (approximately 150 mL size) and heated on a hot plate or other heating device to the temperature of 250°F for type I and 350°F for types II and IV. The product shall be examined for odor at the temperatures specified above, and flavor after cooling to approximately 100°F (see 3.3.1.1 and 3.3.1.2).

4.4.3 Pouring test. Pouring test shall be in accordance with 4.4.3.1, 4.4.3.2, and 4.4.3.3. (Not applicable to 55 and 58 gallon drums.)

4.4.3.1 Equipment.

(a) Sieve (required). U.S. Standard Sieve Series, coarse series with 5/8 inch square openings.

(b) Ring stand (optional). Three legged stand, approximate ring diameter dimensions of 7-5/8 inches outside diameter and 6-1/2 inches inside diameter, with legs approximately 9 inches long.

(c) Vessel (optional). Approximately 10 inches deep and 10 inches in diameter for collecting the shortening which passes through the sieve.

4.4.3.2 Procedure.

(a) Temper the product in the unopened container in a water bath until the temperature of the entire contents of the container is 70 ± 10°F. Temperature at center of container shall be checked by inserting a thermometer through a small opening in end of container.

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(b) Place the ring stand in collecting vessel and place sieve on top of ring stand frame.

(c) Remove end of container in such a manner as to leave no jagged edges.

(d) Pour entire contents of container through sieve, rest container in sieve and allow to drain for 30 seconds. Remove the container from sieve.

4.4.3.3 Results. Observe the sieve and inside of the emptied container. In order to meet the requirements of this test, the contents shall have flowed from the container in such a manner as not to leave any lumps of product remaining on the sieve or clinging to the inside of the container. Not more than 1/8 inch layer of product may remain on the inside of the container (including the lid).

4.4.4 Foam test. The foam test for type IV, classes 1 and 2 product, shall be performed as specified in 4.4.4.1 and 4.4.4.2.

4.4.4.1 Preparation of sample. Four to five pounds of the product to be tested shall be held at a temperature of 440 ± 5°F in a small electric deep fat fryer, or other suitable device, which exposes approximately 55 square inches of the fat to the atmosphere, for 7 hours, on each of two consecutive days. The stressed fat shall then be tested as specified in 4.4.4.2.

4.4.4.2 Conducting the test. A 40-gram sample of the frying fat, prepared as specified in 4.4.4.1, shall be weighed into a flat-bottom pyrex tube (8 inches long by 1-3/4 inches in diameter), and heated over a gas flame or electric hot plate to 395 ± 5°F. The tube shall then be removed from the heat source and six pieces of previously prepared, freshly cut potato, 1-inch long and 3/8 inch in diameter, placed on the end of a wire shall be immersed into the hot fat for 1 minute. The foam at no time shall exceed 2-1/2 inches at its maximum height.

4.4.5 Tin coat test. The tin coating shall be tested in accordance with PPP-C-96.

4.4.6 Leakage test for 5-gallon can and 1-gallon oblong can. The filled and closed can shall be tested for leakage by submerging the can in water in such a manner that the can is lying on a side with not less than 3 inches of water over any portion of the can, and not less than 1 inch of water between the lower portion of the can and the bottom of the tank. The can shall be held in this position for not less than 30 seconds and shall then be rotated so that observation for leaks can be made with each of the four sides facing the top of the tank. The temperature of the water shall be 100 ± 5°F. A leak consists of a steady progression of bubbles. Isolated bubbles caused by entrapped air shall not be considered as a sign of leaks.

4.4.7 Thickness of film and heat seal strength. The thickness of the film and the heat seal strength of the seams shall be tested in accordance with L-P-378.

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4.4.8 Thickness of plastic container. Wall thickness of the plastic container shall be measured with a dial caliper gauge accurate to ± 0.001-inch. The gauge shall be an anvil type used for measurement of cylindrical tubing. Choose a flat area approximately halfway down the container height and make a minimum of twelve measurements at evenly spaced intervals around the perimeter of the container to determine the minimum wall thickness.

4.4.9 Examination of can coating. Examination of can coating shall be performed in accordance with the examination criteria of TT-C-495.

4.4.10 Examination of can labeling. Examination of can labeling shall be performed in accordance with the examination criteria of MIL-L-1497 or regulations of FSIS, as applicable.

4.4.11 Examination of unit loads. Unitized loads shall be examined in accordance with the criteria of MIL-L-35078 and the applicable specification sheet for the type and class of load specified.

5. PACKAGING

5.1 Preservation. The product shall be unit packed in accordance with level A, B, or C, as specified (see 6.2). Net weight or volume shall be as specified in 5.1.1, 5.1.2, or 5.1.3. A minus tolerance, as shown in table VI, will be allowed in any individual container, provided the average net weight or volume of the containers inspected in accordance with table VI is not less than the net weight or volume specified.

5.1.1 Level A.

5.1.1.1 Size 603 x 700 can. Five pounds, 8 ounces of type II product or 5 pounds, 9 ounces of type IV product shall be filled into a size 603 x 700 can in accordance with the level A requirements of PPP-C-29.

5.1.1.1.1 Size 610 x 908 (5-quart) can. Alternatively, 9 pounds, 9 ounces of type III product or 9 pounds, 11 ounces of type IV product shall be filled into a size 610 x 908 (5-quart) can in accordance with the level A requirements of PPP-C-29.

5.1.1.2 Five-gallon square can. Thirty-seven pounds of the type I product or 33 pounds of the type II, class 1 (a and b) or class 2 (a, b, or c) or type III, class 2 (a and b) or type IV, class 2 (a and b) product shall be filled into a square metal can 9-3/8 by 9-3/8 by 13-7/8 or 13-3/4 inches with soldered side seam and double-seamed soldered or squeezed-on soldered top and bottom. Alternatively, the can shall have a welded side seam with compound lined, double-seamed top and bottom ends. The can shall be made throughout from not less than commercial 0.25 pound per base box electrolytic tinplate (see 4.4.5). The ends of the can shall have enameled interiors in accordance with the applicable requirements of PPP-C-29. The can shall have sufficient base plate weight and temper to protect the product during shipment and storage. When specified (see 6.2), the can top may be one of the following styles:

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(a) Equipped with a cover not less than 6-3/16 inches in diameter, compound-lined and double-seamed in position.

(b) Equipped with a cover 6-7/8 inches in diameter, compound-lined and rolled or clinched over a flange opening forming a joint inclining outwards approximately 20 degrees from the vertical.

(c) Fitted in one corner with a 2-inch diameter, compound-lined, Newman style cap or equal, tightly rolled or clinched over the neck of the opening so that after closing, the skirt of the cap is crimped inward to form an angle of at least 10 degrees from the vertical. (For type III, class 2 (b) and type IV, class 2 liquid shortening only).

(d) Fitted in one corner with a flexible spout closure. (For type III, class 2 (b) and type IV, class 2 liquid shortening only).

The covers used shall be made from not less than commercial 0.25 pound per base box electrolytic tinplate (see 4.4.5). The can shall be coated overall on the outside with a coating conforming to type I or, when specified, (see 6.2), type III of TT-C-495. The filled can shall be hermetically sealed and shall not leak when tested in accordance with 4.3.4.2. Stripe coating of soldered side seam is not required.

5.1.1.3 One-gallon oblong can. One gallon of type IV, class 2, product shall be filled into an oblong metal can with soldered or cemented side seam and compound-lined, double-seamed ends in accordance with type V, class 3 or 8, of PPP-C-96. The can shall be made throughout from not less than commercial 0.25 pound per base box electrolytic tinplate, and shall be coated overall with a coating conforming to type I or, when specified (see 6.2), type III of TT-C-495. The filled and sealed can shall not leak when tested in accordance with 4.3.4.2.

5.1.2 Level B.

5.1.2.1 Polyethylene bag in fiberboard box (cube). Fifty pounds of type II, class 1 (b) or type III, class 2 (a) product shall be filled into a bag made from polyethylene film having a minimum thickness of 0.002 inch. The bag shall be heat-sealed along three edges; the seals having a minimum heat-seal strength of 60 percent of the film tensile strength. The bag shall be formed inside the fiberboard container constructed and closed (method II) in accordance with style RSC, grade 200, type CF (variety SW) or SF, class domestic of PPP-B-636. The height of the set-up bag shall be equal to the depth of the container, plus the full width of the container. The top of the bag shall be neatly folded down over the product before closure of the container. Grease seepage or staining caused by seepage shall not be evident on the exterior surface of the completed package.

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5.1.2.2 Cans.

5.1.2.2.1 Sizes 603 x 700, 610 by 908 (5-quart), 1-gallon oblong can and 5-gallon square can. Same as specified in 5.1.1, except that cans with or without an exterior coating will be acceptable, and 5-gallon cans may be fabricated from 107-pound base weight of plate.

5.1.2.2.2 Fifty-pound capacity can. Fifty-pounds of type II, class 1 (a) or (b) or type III, class 2 (a) product shall be filled into a new, sound, full-opening, slip cover, closure style metal can size 1204, 1205, or 1206 by 1404, 1408, 1504, 1505, 1510, or 1512. The cans shall be constructed of CMQ (Can Making Quality) or 0.25-pound tinplate (see 4.4.5) having a base weight of metal of 107 pounds per base box. The can bottom shall be compound-lined and double-seamed in position and the can rim shall be curled outward. The slip cover shall be snug-fitting and shall have a skirt 11/16 inch long with a curled edge. The can shall be coated overall outside with a coating conforming to type I or when specified (see 6.2), type III of TT-C-495. Stripe coating of soldered side seam is not required.

5.1.2.3 Plastic containers. Two and one-half gallons of type IV, class 2 product shall be filled into a molded plastic container. The container shall be molded from a low density, food grade polyethylene and shall have a minimum wall thickness of 0.025 inches. The container shall have an integral handle and an off-center pouring neck which shall be fitted with a threaded plastic screw cap. The cap shall be lined with a paperboard disc that has been treated to make grease-proof. When the cap is completely screwed onto the container, the rim of the neck shall seat firmly against the inner bearing surface of the liner to effect a liquid-tight seal. The filled and closed container shall not leak when tested in accordance with table V.

5.1.3 Level C.

5.1.3.1 Sizes 603 x 700 and 610 x 908 cans. Same as specified in 5.1.1 except that commercial tinplate or electrolytic chromium coated steel (ECCS) plate cans normally used for the product are acceptable and the requirements for exterior coating is not applicable for tinplate cans. The ECCS cans shall be required to have an exterior commercial coating.

5.1.3.2 Five-gallon square and one-gallon oblong cans. Same as specified in 5.1.1, except that commercial tinplate normally used for the product is acceptable and the requirements for exterior coating is not applicable.

5.1.3.3 Polyethylene bag in fiberboard box (cube). Same as specified in 5.1.2.

5.1.3.4 Fifty-pound capacity can. Same as specified in 5.1.2, except that commercial tinplate normally used for the product is acceptable and the requirement for exterior coating is not applicable.

5.1.3.5 Plastic containers. Same as specified in 5.1.2.

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5.1.3.6 Three-pound capacity cans. Three pounds of type II, class 1 (b) product shall be unit packed in cans normally used by the supplier for retail distribution.

5.1.3.7 Fifty-five or 58-gallon capacity metal drums. Four hundred pounds of type IV, class 2 (b) product shall be unit packed in metal drums conforming to the freight classification specified in 5.2.3.

5.2 Packing. The product shall be packed in accordance with level A, B, or C, as specified (see 6.2). The product in 603 x 700 cans shall be packed in accordance with PPP-C-29 for the level specified.

5.2.1 Level A, cans. One 5-gallon can, or six 4-quart or six 5-quart cans arranged 3 in length and 2 in width, shall be packed on end into a snug-fitting fiberboard box constructed, closed, and reinforced in accordance with style RSC, V2s, of PPP-B-636. One-gallon and 5-quart cans shall be separated from each other by full height partitions constructed from fiberboard conforming to W5c of PPP-F-320.

5.2.2 Level B.

5.2.2.1 Cans. Cans of the product, in the number and arrangement specified in 5.2.1, shall be packed on end in a snug-fitting fiberboard box, constructed, closed, and reinforced in accordance with style RSC, V3c, V3s, or V4s of PPP-B-636. One gallon and 5-quart cans shall be separated from each other by full height partitions constructed of W5c of PPP-F-320.

5.2.2.2 Plastic containers. Three plastic containers of product shall be packed on end in a snug-fitting fiberboard box, constructed, closed, and reinforced in accordance with style RSC-L, V3c, V3s, or V4s of PPP-B-636. Each box shall be furnished with a bottom pad not more than 1/8 inch smaller than the length and width of the box. In addition, plastic containers shall be separated from each other by a full height partition. Pads and partitions shall be constructed of the same material as the container.

5.2.2.3 Three-pound capacity can. Twelve cans of product, arranged 4 in length and 3 in width, shall be packed on end in a snug-fitting fiberboard box, constructed, closed and reinforced in accordance with style RSC, grade V3c, V3s or V4s of PPP-B-636.

5.2.2.4 Fifty-pound capacity cans. Fifty-pound capacity cans, unit packed as specified in 5.1.2, do not require overpacking.

5.2.2.5 Nonmetallic reinforcement of shipping containers. When specified (see 6.2 and 6.4), the shipping containers specified in 5.2.2.1 through 5.2.2.3 shall be reinforced with nonmetallic strapping or pressure-sensitive filament reinforced tape in accordance with the appendix of PPP-B-636.

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5.2.3 Level C.

5.2.3.1 Size 610 x 908 cans, 5-gallon square cans, 1-gallon oblong cans, and 3-pound capacity cans. Cans of product shall be packed in fiberboard containers complying with the Uniform Freight Classification or National Motor Freight Classification, as applicable, except that the closure of fiberboard boxes shall be in accordance with method II as specified in the appendix of PPP-B-636. One-gallon and 5-quart cans shall be separated from each other by full height partitions constructed of the same material as the shipping container.

5.2.3.2 Fifty-pound capacity cans. Cans of product, unit packed as specified in 5.1.2, do not require overpacking.

5.2.3.3 Polyethylene bag in fiberboard box. Product, unit packed as specified in 5.1.2, do not require overpacking.

5.2.3.4 Plastic containers. Three plastic containers of product shall be packed in containers complying with the Uniform Freight Classification or National Motor Freight Classification, as applicable, except that the closure of fiberboard boxes shall be in acccordance with method II as specified in the appendix of PPP-B-636. Each box shall be furnished with a bottom pad not more than 1/8 inch smaller than the length and width of the box. In addition containers shall be separated from each other by a full height partition. Pads and partitions shall be constructed of the same material as the box.

5.2.3.5 Fifty-five or 58-gallon capacity metal drums. Product, unit packed as specified, in 5.1.3, does not require overpacking. The metal drums shall comply with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Unit loads. When specified (see 6.2), the product packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified.

5.4 Labeling and marking.

5.4.1 Civil agencies.

5.4.1.1 Unit containers. Any commercial labeling, or additional labeling as specified that complies with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder is acceptable.

5.4.1.2 Shipping containers. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with FED-STD-123 or regulations of FSIS, as applicable.

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5.4.2 Military agencies.

5.4.2.1 Unit containers, except cans. Same as specified in 5.4.1.1.

5.4.2.1.1 Cans. Cans shall be labeled in accordance with MIL-L-1497 or regulations of FSIS, as applicable.

5.4.2.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129. When the unit pack is a polyethylene bag in fiberboard box (cube), the marking shall include. Store in cool dry place at temperature not to exceed 75°F.

5.4.2.3 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Intended use and technical information. Lard, type I is excellent for use in crisp products such as pie crusts and biscuits as it is a superior contributor of tenderness or shortness to these items. It does not, however, possess suitable creaming and emulsification power for satisfactory use in aqueous batters such as cake mixes. Type II general purpose shortening will have good shortening quality when used in bakery goods by producing a soft, velvety, and tender crumb. It will blend well with other dough components and impart a pleasant flavor to the finished product. Classes 2a, b, and c will have an extended storage life before use over class 1 products and will provide extended stability when used in preparation of military ration components. Type III, class 1 products (high monoglyceride level) will produce yeast-raised products of good tenderness and keeping quality with softness in excess of that obtained from class 2 (low monoglyceride level) products. The latter will produce satisfactory loaf and layer cakes and icings. Type IV products will perform satisfactorily in deep fat frying of all types of foods. When product packaged in polyethylene bag in fiberboard box is to be shipped, handled, and stored at temperatures in excess of 75°F for extensive periods of time, it is recommended the product be kept under refrigerated conditions (40-60°F). Prior to use the product should be tempered to ambient temperature.

6.2 Ordering data. Purchasers should select the preferred option permitted herein and include the following information in procurement documents:

(a) Type and class required (see 1.2.1).
(b) Type I, class 1 only. Whether grainy or smooth texture is required. If no stipulation is made, smooth texture requirement will be in effect (see 3.2.1.1 and 3.3.1.1).
(c) If product possesses a characteristic yellow color, the color shall be so stipulated (see table I, footnote 1/).
(d) When in-process examination is not required by civil agencies (see 4.3.2).

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(e) When type III can coating in accordance with TT-C-495 is required (see 5.1.1.2, 5.1.1.3, and 5.1.2.2).
(f) Applicable levels of preservation and packing (see 5.1 and 5.2).
(g) Type and class of unit load when specified (see 5.3).
(h) When nonmetallic reinforcement is required for level B packing (see 5.2.2.5 and 6.4).
(i) When packing specified in 5.2.3.5 is required (see 6.5).

6.3 Military agencies. Based on conditions known or expected to be encountered during shipping, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15, NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.4 Selection of nonmetallic reinforcement. The nonmetallic reinforcement specified in 5.2.2.5 should be selected for fiberboard shipping containers that are destined for transfer at sea or other specific overseas operations.

6.5 Destination inspection (for civil agencies only). When the finished product has been inspected and passed at point other than destination, the contract should require that the product be inspected at destination for condition and quantity only.

MILITARY INTEREST:

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD
Navy - MS, MC

Civil Agency Coordinating Activities:

DOJ - BOP
DOT - CGS
HHS - FDA, NIH
USDA - FQA
VA - OSS

Preparing activity:

Army - GL

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